

If you're preparing for **LeetCode in Java**, you don't need every method in the Java API. You need the methods that are commonly used in DSA.

1. String Methods (`java.lang.String`)

Method	Description	Example
<code>length()</code>	Returns length	<code>"abc".length()</code> → 3
<code>charAt(i)</code>	Character at index	<code>"abc".charAt(1)</code> → 'b'
<code>substring(start)</code>	From index to end	<code>"hello".substring(2)</code> → "llo"
<code>substring(start,end)</code>	Part of string	<code>"hello".substring(1,4)</code> → "ell"
<code>equals(str)</code>	Compare strings	<code>"abc".equals("abc")</code>
<code>equalsIgnoreCase(str)</code>	Ignore case	<code>"ABC".equalsIgnoreCase("abc")</code>
<code>compareTo(str)</code>	Lexicographical comparison	<code>"abc".compareTo("abd")</code>
<code>contains(str)</code>	Check substring	<code>"hello".contains("ell")</code>
<code>startsWith(str)</code>	Prefix check	<code>"apple".startsWith("app")</code>
<code>endsWith(str)</code>	Suffix check	<code>"apple".endsWith("ple")</code>
<code>indexOf(ch)</code>	First occurrence	<code>"banana".indexOf('a')</code>
<code>indexOf(str)</code>	First substring	<code>"banana".indexOf("na")</code>
<code>lastIndexOf(ch)</code>	Last occurrence	<code>"banana".lastIndexOf('a')</code>
<code>isEmpty()</code>	Empty string?	<code>"".isEmpty()</code>
<code>isBlank()</code>	Empty or spaces (Java 11+)	<code>" ".isBlank()</code>
<code>toLowerCase()</code>	Lowercase	<code>"ABC".toLowerCase()</code>
<code>toUpperCase()</code>	Uppercase	<code>"abc".toUpperCase()</code>

Method	Description	Example
<code>trim()</code>	Remove leading/trailing spaces	<code>" hi ".trim()</code>
<code>strip()</code>	Unicode-aware trim	<code>" hi ".strip()</code>
<code>replace(old,new)</code>	Replace chars/string	<code>"aabb".replace('a','c')</code>
<code>replaceAll(regex,new)</code>	Regex replace	<code>"123abc".replaceAll("\\d","")</code> <code>)</code>
<code>split(regex)</code>	Split string	<code>"a,b,c".split(",")</code>
<code>concat(str)</code>	Concatenate	<code>"Hi".concat(" All")</code>
<code>repeat(n)</code>	Repeat string	<code>"ab".repeat(3)</code>
<code>matches(regex)</code>	Regex matching	<code>"123".matches("\\d+")</code>
<code>toCharArray()</code>	Convert to char array	<code>"abc".toCharArray()</code>
<code>intern()</code>	String pool	Rarely used
<code>valueOf(x)</code>	Convert object/primitive to string	<code>String.valueOf(123)</code>
<code>join(delimiter,...)</code>	Join strings	<code>String.join("-",a,b)</code>
<code>format()</code>	Format strings	<code>String.format("%d",10)</code>

2. StringBuilder Methods

Very important for interviews.

</> Java



```
StringBuilder sb = new StringBuilder();
```

Method	Example
<code>append()</code>	<code>sb.append("abc")</code>
<code>insert()</code>	<code>sb.insert(1,"x")</code>

Method	Example
<code>delete()</code>	<code>sb.delete(2,5)</code>
<code>deleteCharAt()</code>	<code>sb.deleteCharAt(0)</code>
<code>replace()</code>	<code>sb.replace(1,3,"hi")</code>
<code>reverse()</code>	<code>sb.reverse()</code>
<code>charAt()</code>	<code>sb.charAt(2)</code>
<code>setCharAt()</code>	<code>sb.setCharAt(1,'a')</code>
<code>length()</code>	<code>sb.length()</code>
<code>capacity()</code>	<code>sb.capacity()</code>
<code>setLength()</code>	<code>sb.setLength(5)</code>
<code>substring()</code>	<code>sb.substring(2)</code>
<code>toString()</code>	Convert to String

3. Character Class Methods (`Character`)

Useful in string questions.

Method	Example
<code>isDigit(c)</code>	<code>'5'</code>
<code>isLetter(c)</code>	<code>'A'</code>
<code>isLetterOrDigit(c)</code>	<code>'A'</code>
<code>isUpperCase(c)</code>	<code>'A'</code>
<code>isLowerCase(c)</code>	<code>'a'</code>
<code>toUpperCase(c)</code>	<code>'a' → 'A'</code>
<code>toLowerCase(c)</code>	<code>'A' → 'a'</code>
<code>isWhitespace(c)</code>	<code>' '</code>

4. Array Methods (`java.util.Arrays`)

Import:

Java



```
import java.util.Arrays;
```

Method	Example
<code>sort(arr)</code>	Ascending sort
<code>parallelSort(arr)</code>	Multi-threaded sort
<code>binarySearch(arr, key)</code>	Search sorted array
<code>fill(arr, val)</code>	Fill entire array
<code>copyOf(arr, n)</code>	Copy first n elements
<code>copyOfRange(arr, a, b)</code>	Copy range
<code>equals(arr1, arr2)</code>	Compare arrays
<code>deepEquals()</code>	Compare 2D arrays
<code>compare()</code>	Lexicographical comparison
<code>mismatch()</code>	First differing index
<code>toString(arr)</code>	Print array
<code>deepToString(arr)</code>	Print 2D array
<code>stream(arr)</code>	Convert to Stream
<code>setAll(arr, ...)</code>	Initialize
<code>parallelSetAll()</code>	Parallel initialization

Example:

Java



```
int[] nums = {5,2,7};

Arrays.sort(nums);

System.out.println(Arrays.toString(nums));
```

Output

```
[2,5,7]
```



5. Array Utility Methods (Manual)

These are not methods on arrays, but operations you'll use constantly.

Length

```
</> Java

arr.length
```



Clone

```
</> Java

int[] copy = arr.clone();
```



Iterate

```
</> Java

for(int num : arr)
```



Convert Array to List

```
</> Java

Arrays.asList(arr)
```



(Works directly only with object arrays like `String[]`, not primitive arrays such as `int[]`.)

6. Common Array Operations

Find Maximum

Java



```
int max = Integer.MIN_VALUE;

for(int x : arr)
    max = Math.max(max, x);
```

Find Minimum

Java



```
int min = Integer.MAX_VALUE;

for(int x : arr)
    min = Math.min(min, x);
```

Reverse Array

Java



```
int left = 0;
int right = arr.length - 1;

while(left < right){
    int temp = arr[left];
    arr[left] = arr[right];
    arr[right] = temp;

    left++;
    right--;
}
```

Sum

Java



```
int sum = 0;
```

```
for(int x : arr)
    sum += x;
```

Copy

</> Java



```
int[] copy = Arrays.copyOf(arr, arr.length);
```

7. Math Methods Frequently Used

Method	Example
<code>Math.max()</code>	<code>Math.max(a,b)</code>
<code>Math.min()</code>	<code>Math.min(a,b)</code>
<code>Math.abs()</code>	Absolute value
<code>Math.pow()</code>	Power
<code>Math.sqrt()</code>	Square root
<code>Math.ceil()</code>	Round up
<code>Math.floor()</code>	Round down
<code>Math.round()</code>	Round nearest
<code>Math.random()</code>	Random number

8. Most Important Methods for LeetCode

These are the ones you'll use in a large percentage of string and array problems.

Strings

- `length()`
- `charAt()`
- `substring()`

- `equals()`
- `contains()`
- `indexOf()`
- `lastIndexOf()`
- `startsWith()`
- `endsWith()`
- `replace()`
- `split()`
- `toCharArray()`
- `toLowerCase()`
- `toUpperCase()`
- `trim()`

StringBuilder

- `append()`
- `delete()`
- `insert()`
- `reverse()`
- `setCharAt()`
- `charAt()`
- `toString()`

Arrays

- `Arrays.sort()`
- `Arrays.binarySearch()`
- `Arrays.fill()`
- `Arrays.copyOf()`
- `Arrays.copyOfRange()`
- `Arrays.equals()`
- `Arrays.toString()`
- `Arrays.deepToString()`
- `arr.length`
- `arr.clone()`

Character

- `Character.isDigit()`
- `Character.isLetter()`
- `Character.isLetterOrDigit()`
- `Character.toUpperCase()`

- `Character.toLowerCase()`

Mastering these methods will cover the vast majority of string and array problems on LeetCode, especially the Easy and Medium difficulty levels.